

DISTRIBUTED LOG AGGREGATION SYSTEM

COURSE: SOCKET PROGRAMMING

NAME: POOJA KOPPAD

SRN: PES2UG24CS352

NIDHI KARKADA

PES2UG24CS314

SHREYAS BELLAD

PES2UG25CS824

Github link –

<https://github.com/Shreyas37-p/CN-mini-project.git>

AIM-

To design and implement a Distributed Log Aggregation System using UDP sockets to collect **encrypted logs** from multiple clients and **decrypt and process them in real time** at a central server.

PROBLEM STATEMENT-

In distributed systems, logs are generated across multiple machines. Managing and analyzing these logs centrally in real time **while ensuring data security** is challenging. This project aims to design a system that aggregates **encrypted logs** from multiple clients and securely processes them at a central server using UDP communication with **decryption at the server side**.

SYSTEM ARCHITECTURE

Client1 —▶

|—▶ Server (Log Aggregator)

Client2 —▶

ALGORITHM-

◆ Server Algorithm

1. Create UDP socket
 2. Bind socket to port
 3. Wait for incoming encrypted messages
 4. Receive encrypted message using `recvfrom()`
 5. Decrypt the received message
 6. Add timestamp
 7. Display decrypted message with log
 8. Repeat
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◆ Client Algorithm

1. Create UDP socket
2. Read input message
3. Encrypt the message
4. Send encrypted message using `sendto()`
5. Repeat

KEY FEATURES

1. **Streaming Ingestion**
Logs are continuously sent from clients to the server in **encrypted form**.
2. **Time Ordering**
Server decrypts the received logs and adds timestamps to each log.
3. **Backpressure Handling**
Server processes decrypted logs sequentially. UDP may drop encrypted packets if overloaded.
4. **Throughput Evaluation**
Number of encrypted logs received and successfully decrypted per second is observed.

SS- Server-

```
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project$ gcc server.c -o server -lssl -lcrypto
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project$ ./server
Secure Log Server Started on UDP port 5000
[2026-04-22 07:35:52] Client1: connection successfull
Logs: 1 | Data: 32 bytes | Throughput: 0.33 bytes/sec
[2026-04-22 07:36:49] Client2: error reporting
Logs: 2 | Data: 64 bytes | Throughput: 0.42 bytes/sec
[2026-04-22 07:36:58] Client2: is connection succesful
Logs: 3 | Data: 112 bytes | Throughput: 0.69 bytes/sec
```

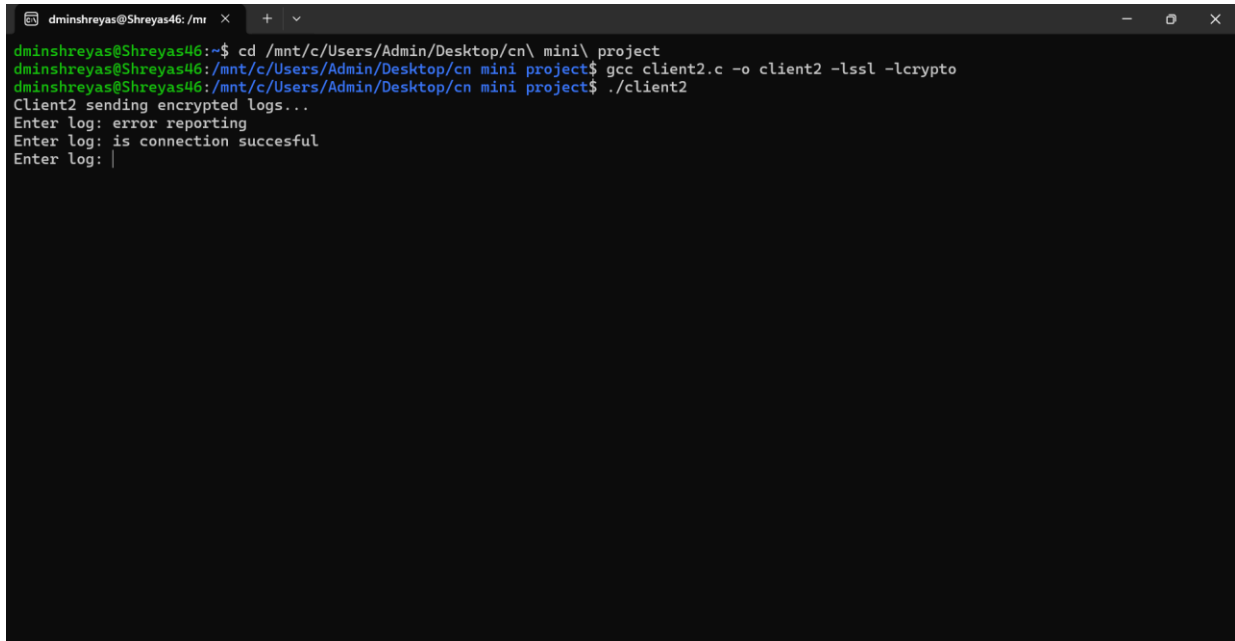
= Server Execution Displaying Decrypted Client Logs with Time Stamps and Performance Metrics.

Client1 –

```
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project$ gcc client1.c -o client1 -lssl -lcrypto
dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project$ ./client1
Client1 sending encrypted logs...
Enter log: connection successfull
Enter log: |
```

= Client1Interface for Inputting and Encrypting Logs Before Transmission.

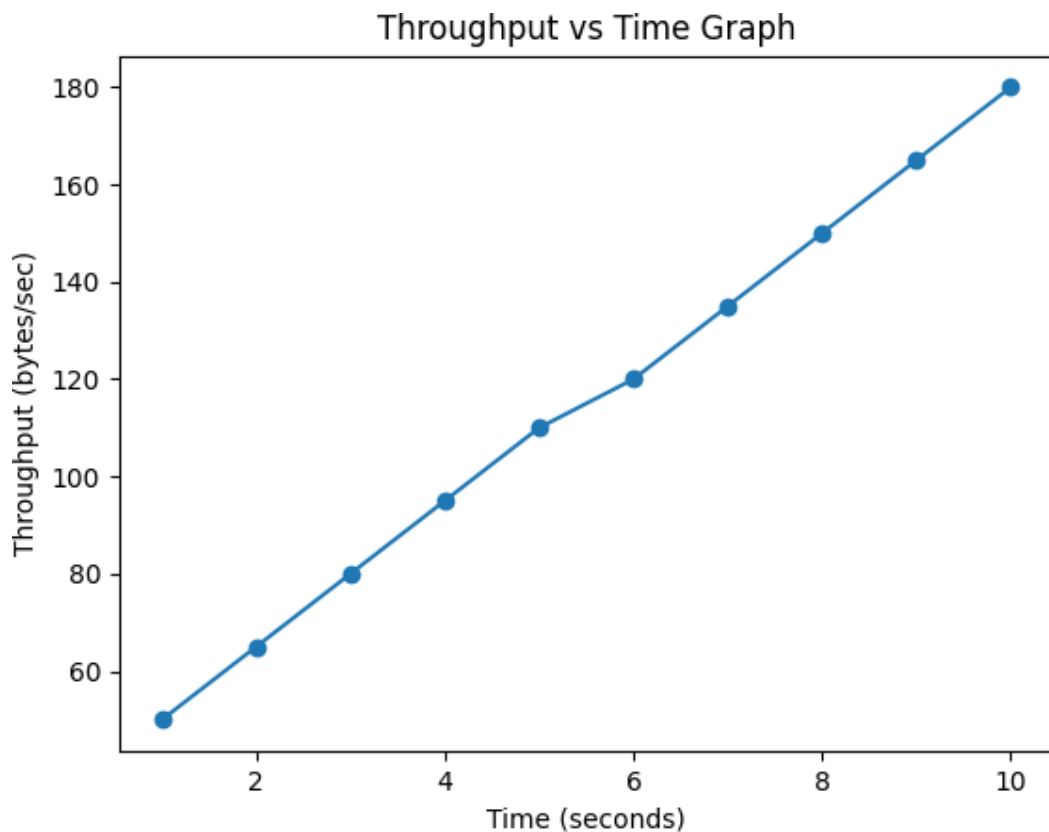
Client2-

A terminal window with a dark background and light green text. The window title is 'dminshreyas@Shreyas46: /mnt/c/Users/Admin/Desktop/cn\ mini\ project'. The terminal shows the following commands and output:

```
dminshreyas@Shreyas46:~$ cd /mnt/c/Users/Admin/Desktop/cn\ mini\ project
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/cn mini project$ gcc client2.c -o client2 -lssl -lcrypto
dminshreyas@Shreyas46:/mnt/c/Users/Admin/Desktop/cn mini project$ ./client2
Client2 sending encrypted logs...
Enter log: error reporting
Enter log: is connection succesful
Enter log: |
```

= Client2 Interface for Inputting and Encrypting Logs Before Transmission.

Graph-



RESULT-

The system successfully collected, encrypted, transmitted, and decrypted logs from multiple clients in real time using UDP communication.

CONCLUSION-

This project demonstrates real-time communication and secure distributed log aggregation using UDP sockets. By integrating encryption at the client side and decryption at the server side, it ensures data confidentiality while maintaining an efficient and lightweight method for collecting logs from multiple sources.